Deep Learning for Economics and Finance

Solving nonlinear dynamic stochastic models in 4 to 100+ dimensions

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Based on: Azinovic, Gaegauf & Scheidegger (2022), *International Economic Review* 63(4), 1471–1525.
Brumm, Krause, Schaab & Scheidegger (2022), "Sparse Grids for Dynamic Economic Models," *Oxford Research Encyclopedia of Economics and Finance*, Oxford University Press.

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Roadmap of This Lecture

Part A

- **Part I.** The IRBC model: preferences, production, TFP, adjustment costs, irreversibility, Fischer–Burmeister
- Equilibrium system + finger exercise

Part B

- **Part II.** DEQN formulation: states, architecture, loss, Gauss–Hermite, training algorithm
- **Part III.** Implementation: Keras code, convergence, policy functions, validation
- Bridge to the production-code DEQN library

Learning objectives

By the end of this lecture you should be able to:

- Set up the International Real Business Cycle model with multi-country state space, capital adjustment costs, and irreversibility constraints
- Derive first-order conditions (Euler equations, consumption sharing) and convert KKT conditions into smooth complementarity functions (Fischer–Burmeister)
- Formulate the DEQN loss function for multi-country DSGE models and train the network on simulated paths
- Implement policy functions in Keras and validate numerical solutions against theoretical steady states
- Extend the framework to 4–100+ dimensional problems and perform sensitivity analysis on model parameters

Part I

The IRBC Model

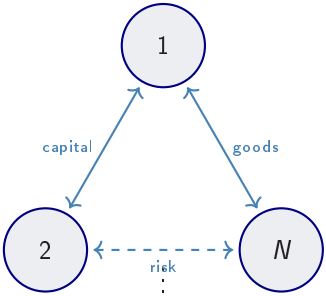
Why International Models?

Key questions in international macro:

- How do capital flows respond to productivity shocks?
- What governs international risk sharing?
- How do trade imbalances emerge?

$\Rightarrow$ Need models with N heterogeneous countries.

$\Rightarrow$ State space: $2N$ dimensions (capital + TFP per country).



Complete markets

Why IRBC for Macro-Finance Research?

Workhorse for open-economy asset pricing and international risk sharing:

- **International risk sharing**. Complete markets imply perfectly correlated consumption; data do not (Backus, Kehoe & Kydland 1992).
- **Market structure & welfare**. Complete vs. incomplete markets; welfare cost of missing insurance (Heathcote & Perri 2002).
- **Capital flows & current account** from intertemporal savings + heterogeneous productivity.
- **Home bias**: frictionless benchmark for observed portfolio holdings.
- **Macro-financial transmission** via adjustment costs, constraints, Pareto weights.

IRBC: Puzzles and Computational Testbed

Macro-finance puzzles:

- Consumption-correlation puzzle (Backus–Kehoe–Kydland 1992): complete markets imply $\rho(c^i, c^j) \approx 1$, but in the data $\rho(c^i, c^j) < \rho(y^i, y^j)$.
- Portfolio home-bias puzzle: optimal diversification predicts large foreign-equity holdings; observed portfolios are strongly domestic.
- Real-exchange-rate (RER) volatility (Backus–Smith): the ratio of price levels P^j/P^i is far more volatile than the consumption ratio c^j/c^i; IRBC with nontraded goods (Stockman & Tesar, 1995) reproduces this.

Computational testbed:

- State dim. linear in N: $2N$.
- Smooth but highly nonlinear.
- Planner gives a clean target.
- Benchmark for projection methods (Krueger & Kubler 2004; Brumm & Scheidegger 2017).

Model Overview

International Real Business Cycle (IRBC) model — Backus, Kehoe & Kydland (1992); here with the DEQN-style extensions of Azinovic, Gaegauf & Scheidegger (2022):

- N countries, each with country-specific capital k^j and TFP z^j
- Complete markets: a social planner allocates resources with Pareto weights τ^j
- Irreversible investment: $I^j \geq 0$ (cannot disinvest)
- Capital adjustment costs: $\Gamma^j = \frac{\kappa}{2} k^j \left(\frac{k^{j\prime}}{k^j} - 1 \right)^2$

Dimensionality

State: $\mathrm{s} = (k^1, \ldots, k^N, z^1, \ldots, z^N) \in \mathbb{R}^{2N}$
Policy: $(k^{1\prime}, \ldots, k^{N\prime}, \lambda, \mu^1, \ldots, \mu^N) \in \mathbb{R}^{2N+1}$

Preferences

Each country j has CRRA utility (with the standard log limit at $\gamma_j = 1$):

$$u^j(c) = \begin{cases} \dfrac{c^{1-1/\gamma_j} - 1}{1 - 1/\gamma_j}, & \gamma_j \neq 1, \\ \ln c, & \gamma_j = 1. \end{cases}$$

- γ_j: intertemporal elasticity of substitution for country j
- Heterogeneous across countries: γ_j linearly spaced in $[\gamma_{\min}, \gamma_{\max}]$
- $\gamma_{\min} = 0.25$, $\gamma_{\max} = 1.0$ (risk aversion $= 1/\gamma_j$ ranges from 1 to 4; the notebook uses the $\ln c$ branch when $\gamma_j = 1$)

Notation reminder

In this chapter γ is the **IES**. Later chapters on continuous-time HA and climate-economics re-purpose γ for CRRA risk aversion (with ψ then standing for the IES). The convention is restated at the top of each chapter.

Social planner maximizes with Pareto weights τ^j:

Production

Each country j produces a single good:

$$Y^j = A_{\text{tfp}} \cdot \exp(z^j) \cdot (k^j)^{\zeta}$$

Calibration of A_{tfp}: chosen so that the steady state is consistent with $k_{ss} = 1$:

$$\boxed{A_{\text{tfp}} = \frac{1 - \beta(1 - \delta)}{\zeta \cdot \beta}}$$

Parameter	Value
ζ (capital share)	0.36
β (discount factor)	0.99
δ (depreciation)	0.01

TFP Process

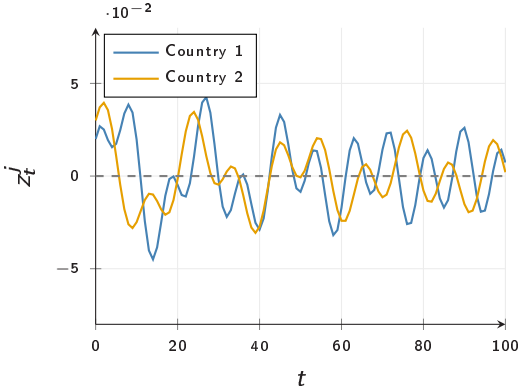
Log TFP follows an AR(1) with common and idiosyncratic shocks:

$$z^{j\prime} = \rho_z z^j + \sigma_e \left(\varepsilon^j + \varepsilon^{\text{agg}}\right)$$

- $\varepsilon^j \sim \mathcal{N}(0,1)$: country-specific
- $\varepsilon^{\text{agg}} \sim \mathcal{N}(0,1)$: aggregate shock
- Total: $N+1$ independent shocks

$\rho_z = 0.95$, $\sigma_e = 0.01$

σ_e is the *per-component* std., so the per-country marginal innovation std. is $\sigma_e\sqrt{2}$ and the cross-country innovation correlation is 1/2 (hard-wired by the equal-weighted decomposition).



Stylized TFP sample paths

Capital Adjustment Costs

Changing the capital stock is costly ($\kappa = 0.5$):

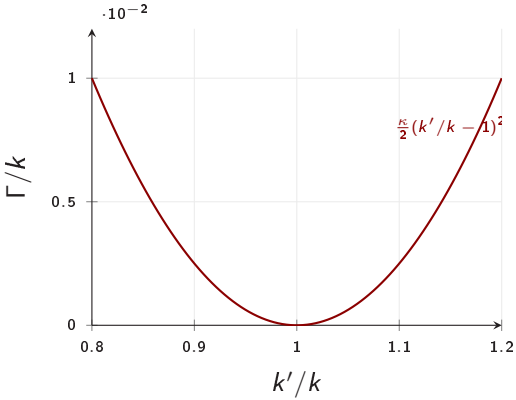
$$\Gamma^j = \frac{\kappa}{2} k^j \left(\frac{k^{j\prime}}{k^j} - 1 \right)^2$$

Key derivatives:

$$\frac{\partial \Gamma}{\partial k'} = \kappa \left(\frac{k'}{k} - 1 \right)$$

$$\frac{\partial \Gamma}{\partial k} = -\frac{\kappa}{2} \left(\frac{k'}{k} - 1 \right) \left(\frac{k'}{k} + 1 \right)$$

Role: smooth investment, break Tobin's $q = 1$, and dampen cross-country capital flows.



Adjustment cost per unit of capital

Putting It Together: The Complete Model

Primitives (from the previous slides):

$$\text{Preferences:} \quad U^j(c) = \frac{c^{1-1/\gamma_j}-1}{1-1/\gamma_j}$$

$$\text{Production:} \quad Y_t^j = A_{\text{tfp}} \exp(z_t^j)\,(k_t^j)^{\zeta}$$

$$\text{TFP process:} \quad z_{t+1}^j = \rho_z\, z_t^j + \sigma_e\left(\varepsilon_{t+1}^j + \varepsilon_{t+1}^{\text{agg}}\right)$$

$$\text{Adjustment cost:} \quad \Gamma_t^j = \frac{\kappa}{2}\, k_t^j \left(k_{t+1}^j / k_t^j - 1\right)^2$$

Constraints:

$$\text{Aggregate resource (ARC):} \quad \sum_{j=1}^{N}\left[Y_t^j + (1-\delta)\, k_t^j - k_{t+1}^j - \Gamma_t^j - c_t^j\right] = 0$$

$$\text{Irreversibility:} \quad I_t^j \equiv k_{t+1}^j - (1-\delta)\, k_t^j \;\geq\; 0 \qquad \forall\, j = 1, \ldots, N$$

The planner chooses $\{c_t^j, k_{t+1}^j\}_{j,t}$ subject to these constraints.

Planner's Problem (Dynamic Optimization)

$$\max_{\{c_t^j,\, k_{t+1}^j\}} \quad \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \sum_{j=1}^{N} \tau^j \, \frac{(c_t^j)^{1-1/\gamma_j} - 1}{1 - 1/\gamma_j} \right]$$

subject to:

$$\text{ARC:} \quad \sum_{j=1}^{N} \left[Y_t^j + (1-\delta)\, k_t^j - k_{t+1}^j - \Gamma_t^j - c_t^j \right] = 0$$

$$\text{Irreversibility:} \quad k_{t+1}^j - (1-\delta)\, k_t^j \;\geq\; 0 \qquad \forall j$$

$$\text{Production:} \quad Y_t^j = A_{\text{tfp}} \exp(z_t^j)\, (k_t^j)^{\zeta}$$

$$\text{TFP:} \quad z_{t+1}^j = \rho_z\, z_t^j + \sigma_e \left(\varepsilon_{t+1}^j + \varepsilon_{t+1}^{\text{agg}} \right)$$

Pareto weights τ^j allocate welfare across countries. We now *attach multipliers* to the constraints to obtain the first-order conditions.

The Lagrangian: Attaching Multipliers

Attach one multiplier per active constraint of the planner's problem:

- λ_t on the aggregate resource constraint — the shadow price of world output at date t.
- $\mu_t^j \geq 0$ on country j's irreversibility — the shadow price of being unable to disinvest.

$$\mathcal{L} = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(\sum_j \tau^j \frac{(c_t^j)^{1-1/\gamma_j} - 1}{1-1/\gamma_j} + \lambda_t \sum_j \left(Y_t^j + (1-\delta)k_t^j - k_{t+1}^j - \Gamma_t^j - c_t^j \right) + \sum_j \mu_t^j \left(k_{t+1}^j - (1-\delta)k_t^j \right) \right) \right]$$

Economic reading:

- λ_t high $\Rightarrow$ world resources scarce $\Rightarrow$ every country cuts consumption.
- $\mu_t^j > 0$ only when country j's irreversibility constraint *binds*.

KKT Conditions for Irreversibility

With the multipliers μ_t^j now defined, optimality of the planner's choice of k_{t+1}^j together with $\mu_t^j \geq 0$ yields the complementary slackness structure:

$$\mu^j \;\geq\; 0, \qquad I^j \;\geq\; 0, \qquad \mu^j \cdot I^j \;=\; 0.$$

Economic reading:

- $I^j > 0$ (agent invests): constraint slack $\Rightarrow \mu^j = 0$.
- $I^j = 0$ (constraint binds): $\mu^j \geq 0$ records the shadow value of enforcing $I^j \geq 0$.

The non-negativity $\mu^j \geq 0$ is the *standard KKT sign*: the multiplier on a $\geq$-constraint is non-negative at the optimum. The smoothed Fischer–Burmeister residual on the next slide encodes this sign restriction *and* the slackness condition $\mu^j I^j = 0$ in one differentiable expression.

Numerical problem: KKT involves min / max — non-differentiable, incompatible with SGD. Next slide: smooth it via the Fischer–Burmeister function.

Fischer–Burmeister Complementarity Function

Problem: KKT conditions involve non-smooth min/max operators.

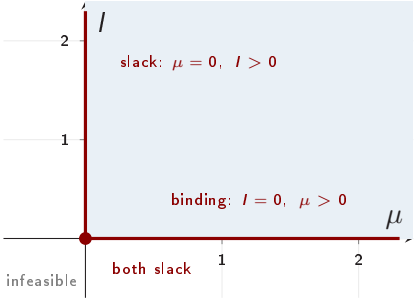
Solution: use a smoothed Fischer–Burmeister (FB) residual:

$$\mathrm{FB}_\varepsilon(\mu, l) = \mu + l - \sqrt{\mu^2 + l^2 + \varepsilon^2}$$

$\varepsilon > 0$ small for numerical stability.

$\mathrm{FB}_0(\mu, l) = 0 \Leftrightarrow \mu \geq 0,\ l \geq 0,\ \mu \cdot l = 0.$
For $\varepsilon > 0$, this exact zero set is slightly smoothed near the origin.

$\Rightarrow$ For $\varepsilon > 0$, differentiable everywhere—compatible with SGD.



Red "L" = zero set of FB. Every point on it satisfies KKT: one of μ, l is zero, the other is ≥ 0. Off the L, $\mathrm{FB} \neq 0$ and the loss penalizes the deviation.

FOC: Consumption Sharing

First-order condition w.r.t. c^j:

$$\tau^j \cdot (c^j)^{-1/\gamma_j} = \lambda$$

where λ is the shadow price of the aggregate resource constraint.

Solving for consumption:

$$\boxed{c^j = \left(\frac{\lambda}{\tau^j}\right)^{-\gamma_j}}$$

- λ is the single policy variable governing consumption allocation
- Higher $\lambda \Rightarrow$ resources are scarcer $\Rightarrow$ lower consumption
- Countries with higher γ_j (lower risk aversion) respond more to λ

FOC: Euler Equation

First-order condition w.r.t. $k^{j\prime}$:

$$\lambda\left(1+\Gamma_{k'}\right)=\beta\,\mathbb{E}\left[\lambda'\cdot\mathrm{MPK}^{j\prime}-(1-\delta)\,\mu^{j\prime}\right]+\mu^{j}$$

where:

- $\Gamma_{k'}=\kappa\left(\frac{k'}{k}-1\right)$ is the marginal adjustment cost
- $\mathrm{MPK}^{j}=1-\delta+\zeta A_{\mathrm{tfp}}\exp(z^{j})\,(k^{j})^{\zeta-1}-\frac{\partial\Gamma}{\partial k}$
- μ^{j} is the irreversibility multiplier
- The term $(1-\delta)\,\mu^{j\prime}$ captures the shadow value if the constraint binds tomorrow

Euler Equation: Relative Error Form

Rearranging the Euler equation into relative error form:

$$\boxed{\frac{\beta\,\mathbb{E}\left[\lambda' \cdot \mathrm{MPK}^{j\prime} - (1-\delta)\,\mu^{j\prime}\right] + \mu^{j}}{\lambda\,(1+\Gamma_{k'})} - 1 = 0}$$

Why relative error?

- All N Euler equations are scale-free
- Residuals can be interpreted as percentage errors in the optimality condition
- A residual of 10^{-3} means the agent is off by 0.1%

Aggregate Resource Constraint

All output must be allocated to consumption, investment, and adjustment costs:

$$\boxed{\sum_{j=1}^{N}\Big[\underbrace{Y^j + (1-\delta)\,k^j}_{\text{resources}} - \underbrace{k^{j\prime}}_{\text{investment}} - \underbrace{\Gamma^j}_{\text{adj. cost}} - \underbrace{c^j}_{\text{consumption}}\Big] = 0}$$

- This is a single equation (not N separate ones)
- Under complete markets, goods flow freely between countries
- Consumption c^j is determined by λ and the Pareto weights

Complementarity Conditions

For each country j, the Fischer–Burmeister condition enforces irreversibility:

$$\mathrm{FB}_\varepsilon(\mu^j, I^j) = \mu^j + I^j - \sqrt{(\mu^j)^2 + (I^j)^2 + \varepsilon^2} = 0$$

where $I^j = k^{j\prime} - (1-\delta)\, k^j$.

Total system:

Equation	**Count**	**Unknown**
Euler equations	N	$k^{1\prime}, \ldots, k^{N\prime}$
Aggregate resource constraint	1	λ
Fischer–Burmeister	N	$\mu^1, \ldots, \mu^N$
Total	$2\mathrm{N} + 1$	$2\mathrm{N} + 1$

Summary: Equilibrium System

Well-posed system: $2N + 1$ equations in $2N + 1$ unknowns.

N	**States**	**Policies**	**Equations**
2	4	5	5
4	8	9	9
10	20	21	21
50	100	101	101
100	200	201	201

- Dimensions grow linearly in N—no combinatorial explosion
- Traditional grid methods: $\mathcal{O}(n^{2N})$ grid points
- DEQNs: a single neural network, trained via SGD

PAUSE — 5 min Finger Exercise: IRBC Euler Equations

Exercise

Consider the 2-country IRBC model. Suppose we double the capital depreciation rate δ from the baseline 0.01 to 0.02.

1. Without running any code: will the steady-state capital stock in each country *increase* or *decrease*? Why?
2. Will the optimal savings rate (investment / output) increase or decrease?
3. Open notebook `lecture_04_01_IRBC_DEQN_smooth.ipynb`, change δ, and verify your prediction.

Solution: IRBC Euler Equations

Answer

1. Steady-state capital decreases *if* A_{tfp} is held fixed at its $\delta = 0.01$ value: the Euler equation implies $r = 1/\beta - 1 + \delta$, so higher δ requires a higher marginal product of capital and (with diminishing returns) a lower k^*. **Caveat:** the script's calibration normalizes $A_{\mathrm{tfp}} = (1/\beta - 1 + \delta)/\zeta$ so that $k_{\mathrm{ss}} = 1$ at the baseline; if you re-derive A_{tfp} at the new δ, the normalized k_{ss} stays at 1 and the change shows up in A_{tfp} itself and in the consumption–investment split. The exercise asks for the A_{tfp}-*fixed* convention.
2. The savings rate increases. More capital depreciates each period, so a larger fraction of output must be reinvested just to maintain the (lower) capital stock.
3. In the notebook: changing δ from 0.01 to 0.02 should show the policy function shifting down (lower capital) and the investment share rising.

Part II

DEQN Formulation for the IRBC

State & Policy Variables

$$\mathbf{s} = (k^1, \ldots, k^N,\ z^1, \ldots, z^N)$$

$2N$ states

input

$$\mathcal{N}_\rho : \mathbb{R}^{2N} \to \mathbb{R}^{2N+1}$$

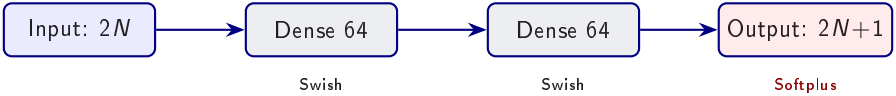
output

$$(k^{1\prime}, \ldots, k^{N\prime},\ \lambda,\ \mu^1, \ldots, \mu^N)$$

$2N + 1$ policies

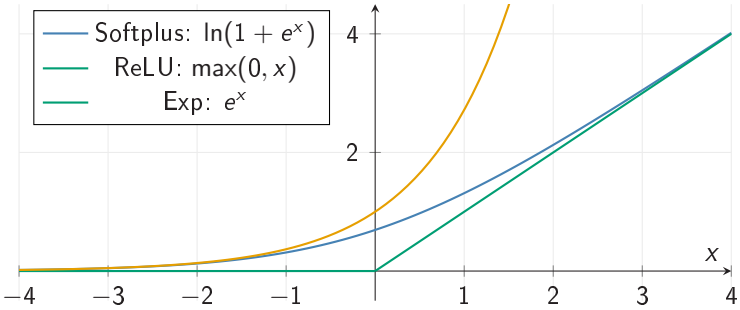
- The neural network maps the full state to all policy variables simultaneously
- For $N = 2$: input dimension 4, output dimension 5
- All outputs must be positive: use softplus activation in output layer

Neural Network Architecture



- **Hidden layers:** 2 layers $\times$ 64 units with Swish activation: $\text{swish}(x) = x \cdot \sigma(x)$
- **Output layer:** Softplus activation: $\ln(1 + e^x) > 0$
- Softplus ensures $k^{j\prime} > 0$, $\lambda > 0$, $\mu^j \geq 0$
- Total parameters: $(2N \cdot 64 + 64) + (64 \cdot 64 + 64) + (64 \cdot (2N + 1) + 2N + 1)$

Why Softplus for Output Activations?



Softplus $\approx$ smooth ReLU: strictly positive, bounded gradients, no overflow risk (unlike e^x).

DEQN Loss Function for the IRBC

The loss is the sum of squared residuals across all equilibrium conditions.

Smooth benchmark (notebook 04a):

$$\boxed{\ell_\rho^{\text{smooth}} = \frac{1}{B}\sum_{i=1}^{B}\left[\sum_{j=1}^{N}\left(\text{Euler}_j^{(i)}\right)^2 + \left(\text{ARC}^{(i)}\right)^2\right]}$$

Irreversibility extension (notebook 04b): add the Fischer–Burmeister block,

$$\ell_\rho^{\text{irrev}} = \ell_\rho^{\text{smooth}} + \frac{1}{B}\sum_{i=1}^{B}\sum_{j=1}^{N}\left(\text{FB}_j^{(i)}\right)^2.$$

where B is the batch size and:

- $\text{Euler}_j = \frac{\beta\,\mathbb{E}[\lambda'\text{MPK}' - (1-\delta)\mu'] + \mu}{\lambda(1+\Gamma_{k'})} - 1 \quad (\text{smooth: } \mu \equiv 0)$
- $\text{ARC} = \sum_j [Y^j + (1-\delta)k^j - k^{j\prime} - \Gamma^j - c^j]$
- $\text{FB}_j = \mu^j + I^j - \sqrt{(\mu^j)^2 + (I^j)^2 + \varepsilon^2} \quad (\textit{04b only})$

$\ell_\rho = 0$ iff all equilibrium conditions hold exactly

Gauss–Hermite Quadrature

The Euler equation requires computing $\mathbb{E}[\cdot]$ over $N+1$ normal shocks.

Gauss–Hermite quadrature in 1D:

$$\mathbb{E}[f(\varepsilon)] = \int f(\varepsilon)\,\phi(\varepsilon)\,d\varepsilon \approx \sum_{q=1}^{Q} w_q\, f(\varepsilon_q)$$

Tensor product for $N+1$ dimensions:

$$\mathbb{E}\Big[f(\varepsilon^1,\ldots,\varepsilon^N,\varepsilon^{\text{agg}})\Big] \approx \sum_{q_1=1}^{Q}\cdots\sum_{q_{N+1}=1}^{Q}\left(\prod_d w_{q_d}\right) f(\varepsilon_{q_1},\ldots,\varepsilon_{q_{N+1}})$$

N	**Shocks**	Q	**Quad. nodes**
2	3	3	27
4	5	3	243
10	11	3	$\sim$ 177K

Computing the Euler Expectation

Step-by-step for each quadrature node q:

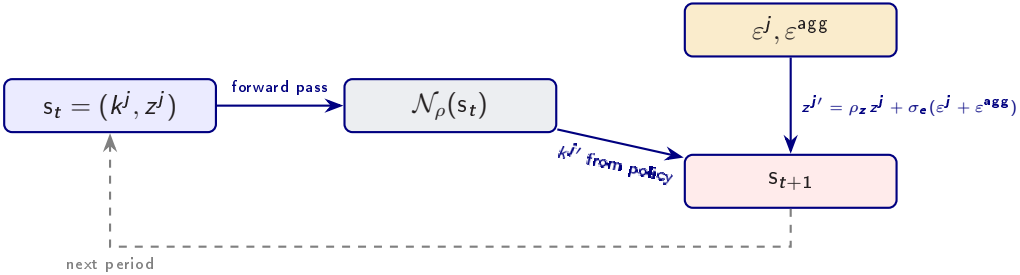
1. **Current state** $\mathsf{s}_t = (k^1, \ldots, k^N, z^1, \ldots, z^N)$
2. **Current policy** $\mathcal{N}_\rho(\mathsf{s}_t) \to (k^{1\prime}, \ldots, k^{N\prime}, \lambda)$ (smooth, $N+1$ outputs; the irreversible extension appends $\mu^1, \ldots, \mu^N$)
3. **Next-period state** at node q:

$$k_q^{j\prime} = k^{j\prime}, \qquad z_q^{j\prime} = \rho_z z^j + \sigma_e(\varepsilon_q^j + \varepsilon_q^{\text{agg}})$$

4. **Next-period policy:** evaluate $\mathcal{N}_\rho(\mathsf{s}_{t+1,q})$
5. **Accumulate:** E-term $= \sum_q w_q \left[\lambda_q' \cdot \text{MPK}_q^j - (1-\delta)\mu_q^{j\prime}\right]$

Each quadrature node requires a forward pass through the network—but all nodes in a batch can be evaluated in parallel.

State Transition



- **Capital:** $k^{j\prime}$ is a policy output (endogenous)
- **TFP:** $z^{j\prime}$ evolves exogenously via AR(1) + shocks
- For simulation: draw $\varepsilon \sim \mathcal{N}(0, 1)$
- For expectations: use quadrature nodes ε_q

Training Algorithm

Algorithm 1: DEQN Training for IRBC (persistent simulation)

Input: Network $\mathcal{N}_\rho$, learning rate η, segments S, monomial nodes/weights
Initialize M trajectory heads $\mathrm{X}^{(1)}, \ldots, \mathrm{X}^{(M)}$ from a feasible starting cloud
for segment $s = 1, \ldots, S$ **do**
  Simulate: for $t = 1, \ldots, T$, sample shocks and step each head one period under $\mathcal{N}_\rho$
  Flatten the $M \cdot T$ recorded states into a training batch
  for pass $p = 1, \ldots, P$ **over mini-batches of the segment do**
    Compute loss ℓ_ρ on the segment (smooth: Euler + ARC; irreversible: also FB)
    Update: $\rho \leftarrow \rho - \eta \nabla_\rho \ell_\rho$ (Adam)
  end for
  Continue: $\mathrm{X}^{(m)} \leftarrow$ terminal state of trajectory m (*no reset to steady state*)
  Monitor: policy drift on `X_anchor`; flag stable when below tolerance
end for
Diagnostic: zero-shock SSS check from dispersed feasible starts
Output: Trained $\mathcal{N}_{\rho^\star}$

Scalability in N

N	States	Policies	NN params	Grid pts ($n=10$)
2	4	5	$\sim$4.8K	10^4
4	8	9	$\sim$5.3K	10^8
10	20	21	$\sim$6.9K	10^{20}
50	100	101	$\sim$17K	10^{100}
100	200	201	$\sim$30K	10^{200}

- NN parameters grow linearly in N (only input/output layers scale)
- Grid methods grow exponentially: 10^{2N} points
- DEQN training cost per step: $\mathcal{O}(|\rho| \cdot B \cdot Q^{N+1})$
- For large N: replace tensor-product GH with monomial rules or QMC
- Theoretical underpinning: deep ReLU networks beat the curse of dimensionality for sparse-grid-representable functions (Montanelli & Du 2019)

From Brock–Mirman to IRBC

	Brock–Mirman	**IRBC**
Countries	1	N
States	(K, z)	$(k^1, \ldots, k^N, z^1, \ldots, z^N)$
Policies	C (or s)	$(k^{1\prime}, \ldots, k^{N\prime}, \lambda, \mu^1, \ldots, \mu^N)$
Equations	1 Euler	N Euler + 1 ARC + N FB
Constraints	none	irreversibility, adj. costs
Shocks	1	$N + 1$
Output act.	sigmoid	softplus
Analytical sol.	yes ($\delta = 1$)	no

The IRBC is a natural scaling test for the DEQN methodology.

Recap: Three Building Blocks

Model

N-country IRBC with irreversible investment, adjustment costs, complete markets. $2N + 1$ equilibrium equations.

Network

$\mathcal{N}_\rho : \mathbb{R}^{2N} \to \mathbb{R}^{2N+1}$ with swish hidden layers, softplus output. All policies in one forward pass.

Training

Persistent simulation: one continuing trajectory ensemble, Adam on the sum of squared residuals.

$\Rightarrow$ **Next:** implementation in TensorFlow/Keras and results.

Part III

Implementation & Results

Notebook Overview

Three notebooks accompany this lecture (two in the L04 `code/` folder, one in the L05 folder):

1. `lecture_04_01_IRBC_DEQN_smooth.ipynb` – *smooth benchmark, in-class walk-through (this lecture's `code/`)*
   - Parameters & steady state, monomial quadrature
   - Keras network, production/consumption/adjustment-cost equations
   - Cost function (Euler + ARC), persistent-simulation training loop
   - Time-invariance check and zero-shock stochastic steady state
2. `lecture_04_02_IRBC_DEQN_irreversible.ipynb` – *irreversible-investment extension (this lecture's `code/`)*
   - KKT multipliers μ^j and feasibility-by-construction policy
   - Fischer–Burmeister complementarity loss
   - Same training loop and diagnostics as the smooth benchmark
3. `lecture_05_05_IRBC_Exercise.ipynb` – *hands-on exercise (lives in L05's `code/`, extends naturally into the NAS / loss-normalization material of Chapter 4)*
   - Task 1: comparative statics on the deterministic steady state

Parameters

Symbol	Name	Value	Description
β	Discount factor	0.99	Quarterly calibration
ζ	Capital share	0.36	Cobb–Douglas
δ	Depreciation	0.01	Low depreciation
ρ_z	TFP persistence	0.95	Highly persistent
σ_e	Shock std. dev.	0.01	Small shocks
κ	Adjustment cost	0.50	Moderate costs
γ_{min}	Min IES	0.25	High risk aversion
γ_{max}	Max IES	1.00	Log utility
k_{ss}	Steady-state capital	1.00	Normalization

Pareto weights: $\tau^j = (A_{\text{tfp}} - \delta)^{1/\gamma_j} = c_{ss}^{1/\gamma_j}$. At the symmetric steady state $c_{ss} = A_{\text{tfp}} - \delta$ for every country, so $\lambda_{ss} = \tau^j (c_{ss})^{-1/\gamma_j} = 1$ shared across all j regardless of heterogeneous γ_j.

Steady State

At steady state ($z^j = 0$, $k^j = k_{ss} = 1$, $\mu^j = 0$):

$$A_{\text{tfp}} = \frac{1 - \beta(1-\delta)}{\zeta\beta} = \frac{1 - 0.99 \cdot 0.99}{0.36 \cdot 0.99} \approx 0.0559$$

$$Y_{ss} = A_{\text{tfp}} \cdot k_{ss}^{\zeta} = A_{\text{tfp}} \approx 0.0559$$

$$I_{ss} = \delta \cdot k_{ss} = 0.01$$

$$c_{ss} = Y_{ss} - I_{ss} \approx 0.0459$$

Verification: the ARC should be satisfied: $\sum_j [Y_{ss} - I_{ss} - c_{ss}] = 0$ ✓

The Euler equation at steady state gives the MPK $= 1/\beta$, which pins down A_{tfp}.

Network Definition (Keras)

```
N_COUNTRIES = 2
n_states = 2 * N_COUNTRIES        # = 4
n_policies = 2 * N_COUNTRIES + 1  # = 5

nn = keras.Sequential([
    keras.layers.Dense(64, activation='swish',
                       input_shape=(n_states,)),
    keras.layers.Dense(64, activation='swish'),
    keras.layers.Dense(n_policies,
                       activation='softplus')
])
```

- swish: $x \cdot \sigma(x)$ — smooth, non-monotone, good gradients
- softplus: $\ln(1 + e^x)$ — ensures all outputs > 0
- Output: $[k^{1\prime}, k^{2\prime}, \lambda, \mu^1, \mu^2]$

Production & Consumption

Production:

```
def production(k, z):
    return A_tfp * tf.exp(z) \
           * k**zeta
```

$$Y^j = A_{\text{tfp}} \exp(z^j)(k^j)^{\zeta}$$

Consumption (from FOC):

```
def consumption(lamb, j):
    tau = (A_tfp - delta)**(1./
        gammas[j])
    return (lamb/tau)**(-gammas[j
        ])
```

$$c^j = (\lambda/\tau^j)^{-\gamma_j}$$

All functions are vectorized over the batch dimension using TensorFlow operations.

Adjustment Costs

```
def adj_cost(k, kp):
    j = kp / k - 1.0
    return 0.5 * kappa * j**2 * k

def adj_cost_kp(k, kp):
    return kappa * (kp / k - 1.0)

def adj_cost_k(k, kp):
    j = kp / k - 1.0
    return -0.5 * kappa * j * (j + 2.0)
```

Marginal product of capital:

```
def mpk(k, z, kp):
    Fk = zeta * A_tfp * tf.exp(z) \
         * k**(zeta - 1)
    return 1 - delta + Fk - adj_cost_k(k, kp)
```

Fischer–Burmeister Function

```
def fischer_burmeister(mu, I, eps=1e-4):
    return mu + I - tf.sqrt(mu**2 + I**2 + eps**2)
```

Investment:

```
def investment(k, kp):
    return kp - (1.0 - delta) * k
```

The FB function is:

- Differentiable everywhere for $\varepsilon > 0$ (unlike $\min(\mu, I)$)
- Exact zero set at $\varepsilon = 0$: $\mu \geq 0$, $I \geq 0$, $\mu \cdot I = 0$
- The small ε prevents division-by-zero at the origin

Cost Function Structure

```
@tf.function
def compute_cost(states, nn):
    policies = nn(states)
    # Extract k', lambda, mu per country
    # For each GH quadrature node:
    #   Compute next state, evaluate nn,
    #   accumulate Euler integrand
    # Euler residual (relative error)
    # ARC: sum resources - uses = 0
    # FB: complementarity per country
    loss = mean(euler**2 + arc**2 + fb**2)
    return loss
```

Key point: the entire computation graph is inside `@tf.function` for automatic differentiation via `GradientTape`.

Persistent-Simulation Training

The companion notebooks train on one continuing trajectory ensemble – no Phase 1 / Phase 2 switch, no resets to steady state.

- Maintain $M = \texttt{N_TRAJECTORIES}$ stochastic trajectory heads $\mathrm{X}^{(1)}, \ldots, \mathrm{X}^{(M)}$.
- Each training segment: simulate the heads forward for $T = \texttt{SIMULATION_LENGTH}$ stochastic periods under the *current* policy network; flatten into a training cloud of size $M \cdot T$; perform a fixed number of SGD passes.
- Continue from the segment's terminal states: $\mathrm{X}_{\mathrm{start}} \leftarrow \mathrm{X}_{\mathrm{end}}$. The trajectory ensemble co-evolves with the policy.
- A `SAMPLING_MODE` switch (`simulation` vs `exogenous`) is available for ablations; the default `simulation` mode never sees a uniform burn-in.

Why this works without a separate burn-in.

- The policy parameterization is feasibility-bounded by construction:
  - smooth (04a): $k^{j\prime} = k^j \exp\{\bar{g} \tanh r_j(\mathrm{s})\}$ keeps $k^{j\prime} > 0$;
  - irreversible (04b): $k^{j\prime} = (1-\delta)k^j + I^j$ with $I^j \geq 0$ enforced via a sigmoid head.
- Random initial weights cannot drive the simulation into infeasible states.

Diagnostics: Time-Invariance and Zero-Shock SSS

Time invariance. The architecture has no calendar-time input, so any fixed weight vector defines a stationary recursive policy. The *numerical* question is whether SGD has stopped moving the policy function.

- Evaluate the policy on a fixed anchor cloud `X_anchor` after each monitoring interval.
- Report `policy_drift_rms` and `policy_drift_max`.
- When both fall below `TIME_INVARIANCE_TOL_RMS` / `_MAX`, the run is flagged stable.

Zero-shock stochastic steady state (SSS). Iterate the learned policy from `ZERO_SHOCK_N_STARTS` dispersed feasible starts with $\varepsilon_t^j = \varepsilon_t^{\text{agg}} = 0$.

- All paths should converge to a common point.
- In a sensible SSS: $I^j \approx \delta\, k^j$ (investment replaces depreciation) and, in 04b, the irreversibility multiplier $\mu^j \approx 0$.

Persistent-Simulation Training: What to Monitor

With a single training pipeline (no Phase 1/Phase 2 switch), *four* signals carry the whole convergence story:

1. **Total loss** decays monotonically across training segments (the *loss-vs-segment* panel in the notebooks).
2. **Per-residual mean magnitudes** – `mean_abs_euler`, `mean_abs_arc`, and (in 04b) `mean_abs_fb` – all fall below the prescribed tolerance.
3. **Policy drift on the anchor cloud** crosses from moving to stable: only then is the policy time-invariant in the recursive sense.
4. **Zero-shock SSS** converges from dispersed feasible starts to a common point with $I^j \to \delta\, k^j$ and (in 04b) $\mu^j \to 0$.

Historical note. An earlier four-approach comparison (uniform vs simulation-only vs two-phase vs ReLoBRaLo) on the same model is preserved in `figures/irbc_4approach_loss.pdf`; it is *not* produced by the current notebooks, which run a single persistent-simulation pipeline.

Convergence: Per-Equation Residuals

After training, we evaluate residuals on a test cloud (smooth + irreversible):

Residual	**Mean** $\lvert\cdot\rvert$	**Max** $\lvert\cdot\rvert$
Euler (country 1)	$\sim 10^{-4}$	$\sim 10^{-3}$
Euler (country 2)	$\sim 10^{-4}$	$\sim 10^{-3}$
ARC	$\sim 10^{-4}$	$\sim 10^{-3}$
Irreversibility extension (notebook 04b only):		
FB (country 1)	$\sim 10^{-5}$	$\sim 10^{-4}$
FB (country 2)	$\sim 10^{-5}$	$\sim 10^{-4}$

- Euler residuals $< 10^{-3} \Rightarrow$ policy is accurate to within 0.1%
- FB residuals near zero $\Rightarrow$ complementarity is well satisfied (04b only)
- These are relative errors—interpretable as percentage deviations

Policy Functions

After training, we visualize the learned policies:

- **Capital policy** $k^{j\prime}$ vs. k^j:
  - Near 45-degree line (close to replacement)
  - Higher investment when TFP is high
- **Shadow price** λ vs. z^j:
  - Decreasing in productivity (more output $\Rightarrow$ lower scarcity)
- **Multiplier** μ^j vs. k^j:
  - Near zero when k is low (investment is positive)
  - Positive when k is high (irreversibility binds)
- **Investment** I^j vs. k^j:
  - Positive most of the time
  - Occasionally zero (constraint active)

Economic Diagnostics

Consumption sharing:

- Under complete markets, c^1 and c^2 should be positively correlated
- Perfect risk sharing: c^1/c^2 is constant
- With heterogeneous γ: ratio varies with λ

Reconnects with the consumption-correlation puzzle flagged on slide 7 of this deck: complete markets predict $\rho(c^1, c^2) \approx 1$, but the data give $\rho(c^1, c^2) < \rho(y^1, y^2)$. Heterogeneous IES alone does *not* resolve the puzzle; closing it with the data needs incomplete markets, frictions, or non-traded goods.

Ergodic distribution:

- Scatter plot of (k^1, k^2) over long simulation
- Should concentrate near (k_{ss}, k_{ss})
- Correlation structure reveals risk-sharing patterns

Complementarity check:

- Plot μ^j vs. l^j
- Should lie on the axes: $\mu = 0$ or $l = 0$
- Fraction of time constraint binds

Validation: Euler LHS vs. RHS

Key diagnostic: compare both sides of the Euler equation on a test set.

LHS: $\lambda \cdot (1 + \Gamma_{k'})$

RHS: $\beta\,\mathbb{E}\big[\lambda' \cdot \mathrm{MPK}' - (1-\delta)\mu'\big] + \mu$

Targets:

- Points should lie on the 45-degree line
- Mean relative error $< 10^{-3}$
- Max relative error $< 10^{-2}$

Also check:

- ARC residual ≈ 0 across all states
- $\mu^j \approx 0$ whenever $I^j > 0$ (complementarity)
- $I^j \geq 0$ everywhere (irreversibility)

These diagnostics confirm the network has learned a valid approximate equilibrium.

Bridge: AutoDiff, Sequence Space, OLG

What we leave with today. An IRBC DEQN trained with a single persistent-simulation pipeline and equal loss weights: the same recipe scales from $N = 2$ to $N = 10$ countries with no architectural change. NAS-tuned width and ReLoBRaLo-balanced loss components are optional refinements, introduced with the NAS / loss-normalization material.

In-class follow-up.

- Reverse-mode automatic differentiation *inside* the equilibrium loss (forward/reverse modes, JVP/VJP, AD pitfalls).
- *Sequence-space* DEQNs: collapsing the recursion into a finite-horizon DAG (Brock–Mirman warm-up, then IRBC).
- OLG with DEQNs: the analytic 6-cohort benchmark of Krueger–Kübler (2004) and the 56-cohort benchmark of Azinovic, Gaegauf & Scheidegger (2022).

Self-study extension.

- Heterogeneous-agent solvers via Young's method (continuum approximated as a histogram).
- Continuum-of-agents DEQN.
- Krusell–Smith with deep learning.

Questions?

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https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Next up:

AutoDiff $\rightarrow$ Sequence Space $\rightarrow$ OLG

(Heterogeneous Agents = self-study)